

WasteZero: Smart Waste Pickup & Recycling Platform

Project Statement: WasteZero is a digital platform to help users schedule waste pickups, categorize recyclables, and promote responsible waste management. Pickup agents are assigned intelligently based on location.

Outcomes:

1. Users can register, login and schedule waste pickups.
2. Waste is categorized (plastic, organic, e-waste, etc.).
3. Agents are notified and assigned dynamically.
4. Users receive notifications and track their waste statistics.
5. Admins monitor overall platform operations.

Modules:

Module A: User Management

Module B: Opportunity Management

Module C: Matching & Communication

Module D: Administration & Reporting

Milestone 1: Weeks 1-2 - User Management:

- Develop user registration and authentication features with role-based access control.
- Implement user profile management, including edit functionality.
- Configure and integrate the backend database to store user information securely.

Output:

- Functional login and registration interfaces.
- User dashboard tailored to each role type.

Milestone 2: Weeks 3-4 - Opportunity Management:

- Enable administrators (NGOs) to create, update, and delete volunteering opportunities.
- Include essential details in each opportunity, such as required skills, duration, and a detailed description.
- Design and implement a database schema to manage opportunity records.

Output:

- Opportunity creation and editing form.
- Opportunity listing interface for users.

Milestone 3: Weeks 5-6 - Matching & Communication:

- Implement a matching algorithm to connect volunteers with opportunities based on Waste Types and geographic location.
- Develop real-time messaging functionality using WebSockets for seamless communication between users.

- Integrate notification system to alert users of new matches and incoming messages.

Output:

- Interactive chat interface for real-time communication.
- Match suggestion dashboard displaying relevant opportunities.

Milestone 4: Weeks 7-8 - Reporting and Administration:

- Develop an admin dashboard to monitor user activity and platform engagement.
- Generate analytical reports on active users, posted opportunities, and volunteer responses.
- Implement administrative controls to suspend users or remove posts when necessary.
- Complete testing and report.

Output:

- Comprehensive admin panel with activity oversight tools.
- Downloadable reports for performance.

Database Schema:

1. Users

- **id**: Unique identifier for the user
- **name**: Full name of the user
- **email**: Email address
- **password**: Hashed password
- **role**: Specifies the role (**volunteer** or **NGO**)
- **skills**: Array of user skills
- **location**: Geographic location
- **bio**: Short biography or profile description

2. Opportunities

- **id**: Unique identifier for the opportunity
- **ngo_id**: Reference to the NGO that created the opportunity (links to **Users.id**)
- **title**: Opportunity title
- **description**: Detailed description of the opportunity
- **required_skills**: Array of skills required for the opportunity
- **duration**: Time commitment or duration of the opportunity
- **location**: Opportunity location
- **status**: Current status (e.g., open, closed, in-progress)

3. Applications

- **id**: Unique identifier for the application
- **opportunity_id**: Reference to the opportunity applied for
- **volunteer_id**: Reference to the volunteer who applied
- **status**: Application status (**pending**, **accepted**, **rejected**)

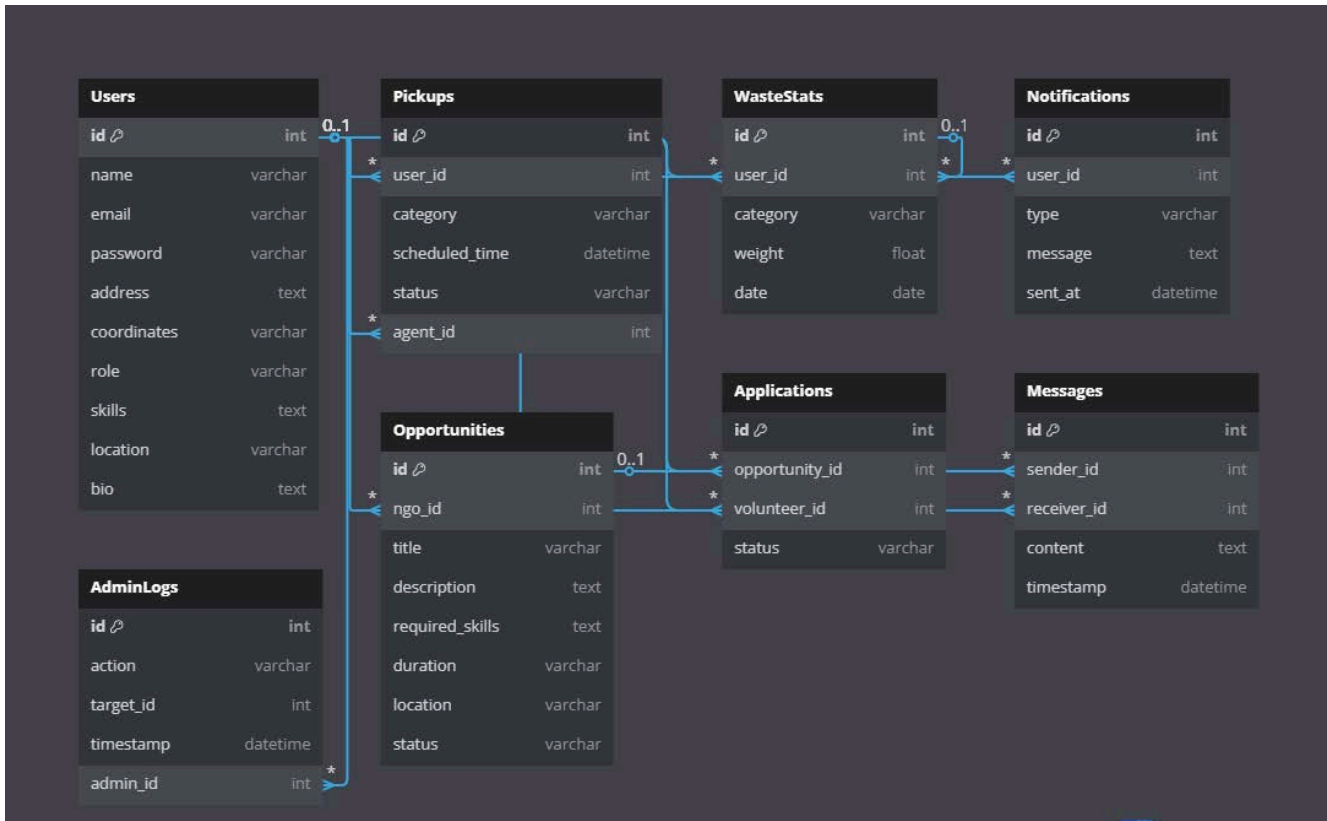
4. Messages

- **id**: Unique message identifier
- **sender_id**: ID of the user who sent the message
- **receiver_id**: ID of the user who received the message
- **content**: Message text
- **timestamp**: Date and time the message was sent

5. AdminLogs

- **id**: Unique identifier for the log entry
- **action**: Description of the admin action performed
- **user_id**: Reference to the user associated with the action
- **timestamp**: Date and time the action was logged

DBdiagram:



Mockups:

Login Page: Milestone 1

 **WasteZero**

Join the Recycling Revolution

WasteZero connects volunteers, NGOs, and administrators to schedule pickups, manage recycling opportunities, and make a positive impact on our environment.

Schedule Pickups

Easily arrange waste collection

Track Impact

Monitor your environmental contribution

Volunteer

Join recycling initiatives

LoginRegister

Login to your account

Enter your credentials to access your account

Username

Your username

Password

Your password

Login

SignUp Page: Milestone 1

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Join recycling initiatives

LoginRegister

Create a new account

Fill in your details to join WasteZero

Full Name

Your full name

Email

Your email

Username

Choose a username

Password

Create a password

Confirm Password

Confirm your password

Role

Volunteer

✓ Volunteer

NGO

Admin

Create Account

MyProfile Page: Milestone 1

WasteZero

venu

Admin

MAIN MENU

Dashboard

Schedule Pickup

Opportunities

Messages

My Impact

SETTINGS

My Profile

Settings

Help & Support

Admin Panel

Dark Mode

Search pickups, opportunities...

My Profile

Manage your account information and settings

Profile

Password

Personal Information

Update your personal information and profile details

Full Name

venu

Email

abc+1@gmail.com

This is the email address used for account notifications.

Location

hydrabad

This helps match you with nearby opportunities.

Skills

WasteZero

ganesh

Volunteer

MAIN MENU

Dashboard

Schedule Pickup

Opportunities

Messages

My Impact

SETTINGS

My Profile

Settings

Help & Support

Dark Mode

Sign Out

Search pickups, opportunities...

My Profile

Manage your account information and settings

Profile

Password

Change Password

Update your password to secure your account

Current Password

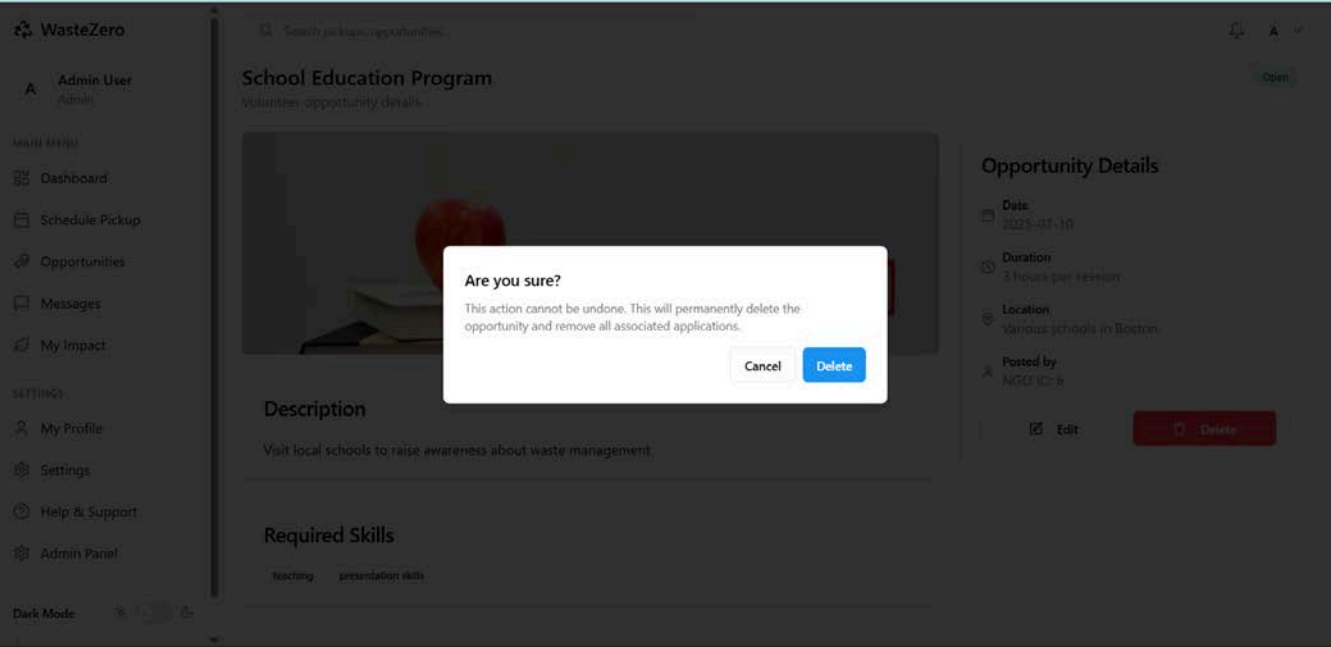
New Password

Password must be at least 6 characters long.

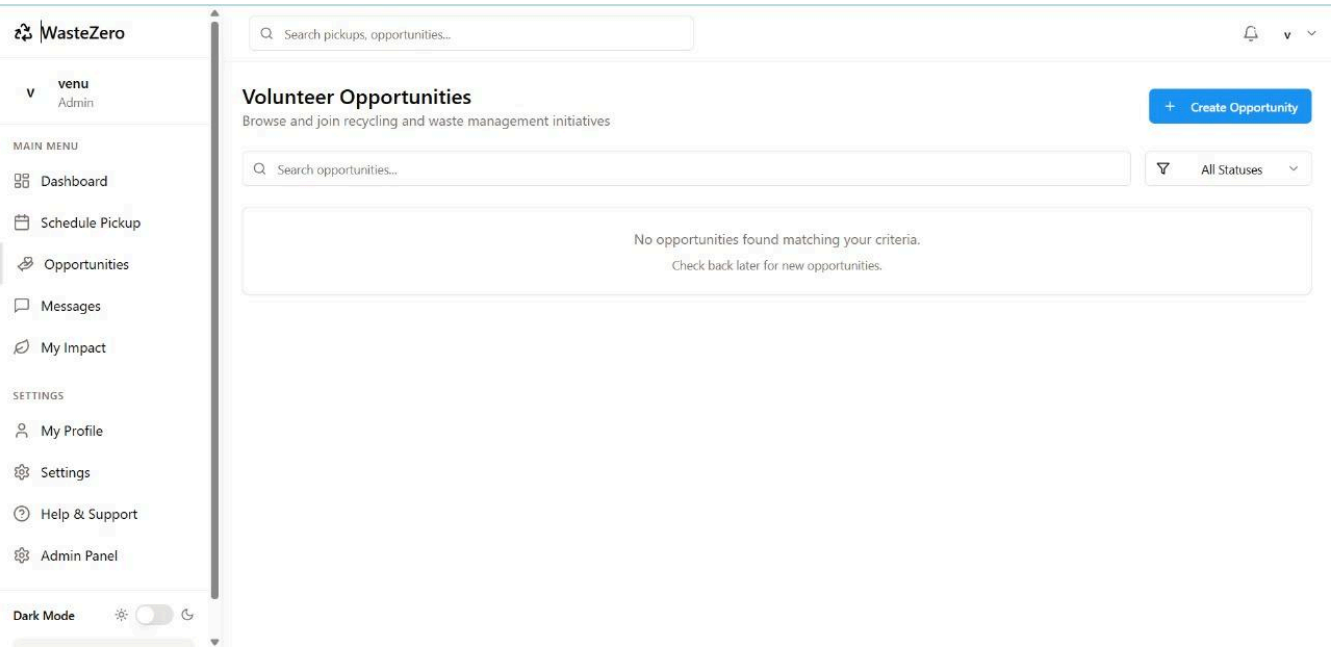
Confirm New Password

Change Password

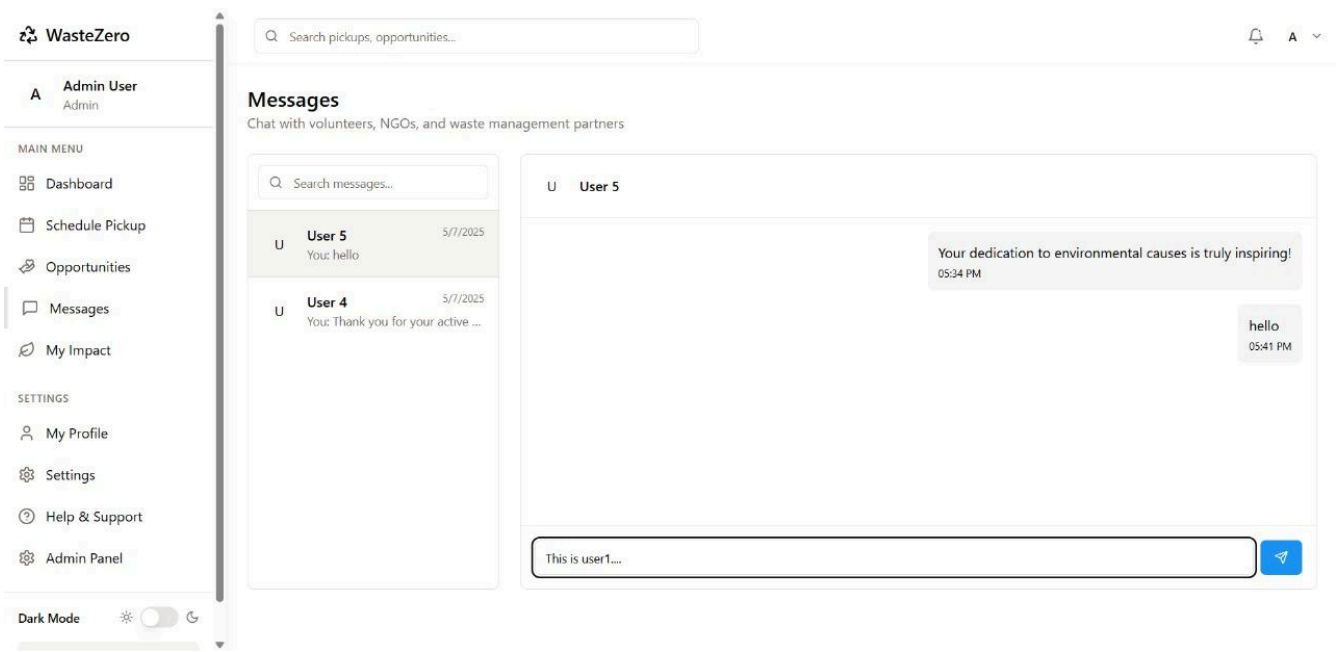
To delete volunteering opportunities:



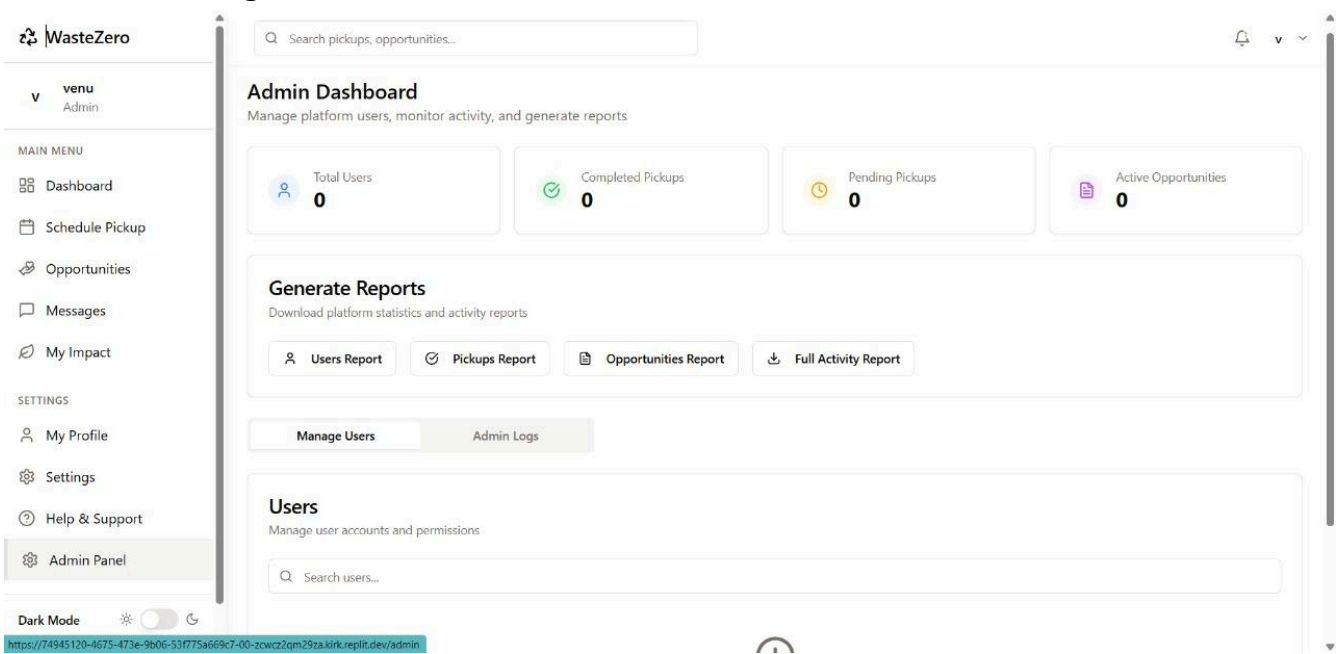
Opportunity Page:



Messages Page: Milestone - 3



Admin Panel Page: Milestone - 4



Dashboard Page: Milestone - 4

WasteZero

A

Admin User

Admin

MAIN MENU

Dashboard

Schedule Pickup

Opportunities

Messages

My Impact

SETTINGS

My Profile

Settings

Help & Support

Admin Panel

Dark Mode

Q

Search pickups, opportunities...

🔔

A

▼

Dashboard

Welcome back, Admin User! Here's your waste management overview.

Total Pickups

28

↑ 7.2% from last month

Recycled Items

635

↑ 12.4% from last month

CO2 Saved (kg)

243

↑ 18.3% from last month

Volunteer Hours

87

↓ 3.1% from last month

Upcoming Pickups

View All

Date & Time

Address

Status

Action

No upcoming pickups scheduled.

Recycling Breakdown

This Month

Plastic

40%

Paper

25%

Glass

15%

E-Waste

10%

Organic

10%

Total Collected

124.5 kg

↑ 10.2%

WasteZero

A

Admin User

Admin

MAIN MENU

Dashboard

Schedule Pickup

Opportunities

Messages

My Impact

SETTINGS

My Profile

Settings

Help & Support

Admin Panel

Dark Mode

Q

Search pickups, opportunities...

🔔

A

▼

Volunteer Opportunities

View All



Beach Cleanup Drive

Open

Join us for a day of cleaning up the shoreline and protecting marine life

📅 2025-06-20

📍 Brighton Beach, Boston

Apply Now



Recycling Workshop

Open

Teach community members about proper recycling techniques

📅 2025-06-15

📍 Community Center, Seattle

Apply Now



School Education Program

Open

Visit local schools to raise awareness about waste management

📅 2025-07-10

📍 Various schools in Boston

Apply Now

Recent Messages

View All

A

Admin User

hello

05:41 PM

A

Admin User

Your dedication to environmental causes is truly inspiring!

05:34 PM

SchedulePickup Page: Milestone - 3,4

WasteZero

venu

Admin

MAIN MENU

Dashboard

Schedule Pickup

Opportunities

Messages

My Impact

SETTINGS

My Profile

Settings

Help & Support

Admin Panel

Dark Mode

Search pickups, opportunities...

v

Schedule Pickup

Request waste collection and manage your pickups

Schedule New Pickup

Pickup History

Request Waste Collection

Fill in the details to schedule a pickup for your recyclable waste

Address

Enter your street address

City

Enter your city

Pickup Date

mm/dd/yyyy

Preferred Time Slot

Select a time slot

Next Step

https://74345120-4675-473e-9b06-53f775a669c7-00-zwcwz2qm29za.kirk.replit.dev/schedule

WasteZero

Admin User

Admin

MAIN MENU

Dashboard

Schedule Pickup

Opportunities

Messages

My Impact

SETTINGS

My Profile

Settings

Help & Support

Admin Panel

Dark Mode

Search pickups, opportunities...

A

Schedule Pickup

Request waste collection and manage your pickups

Schedule New Pickup

Pickup History

Request Waste Collection

Fill in the details to schedule a pickup for your recyclable waste

Waste Types

Select the types of waste you want to recycle

Plastic

Glass

Electronic Waste

Other

Paper

Metal

Organic Waste

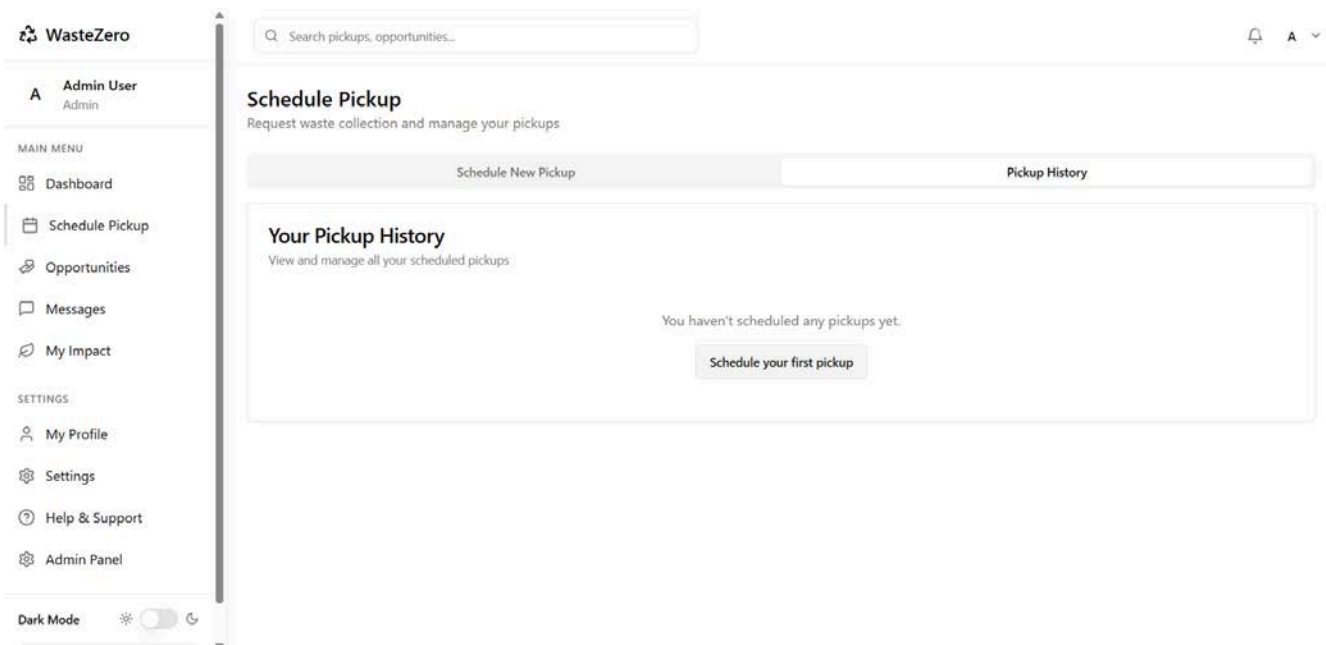
Additional Notes

Any special instructions or information about your waste

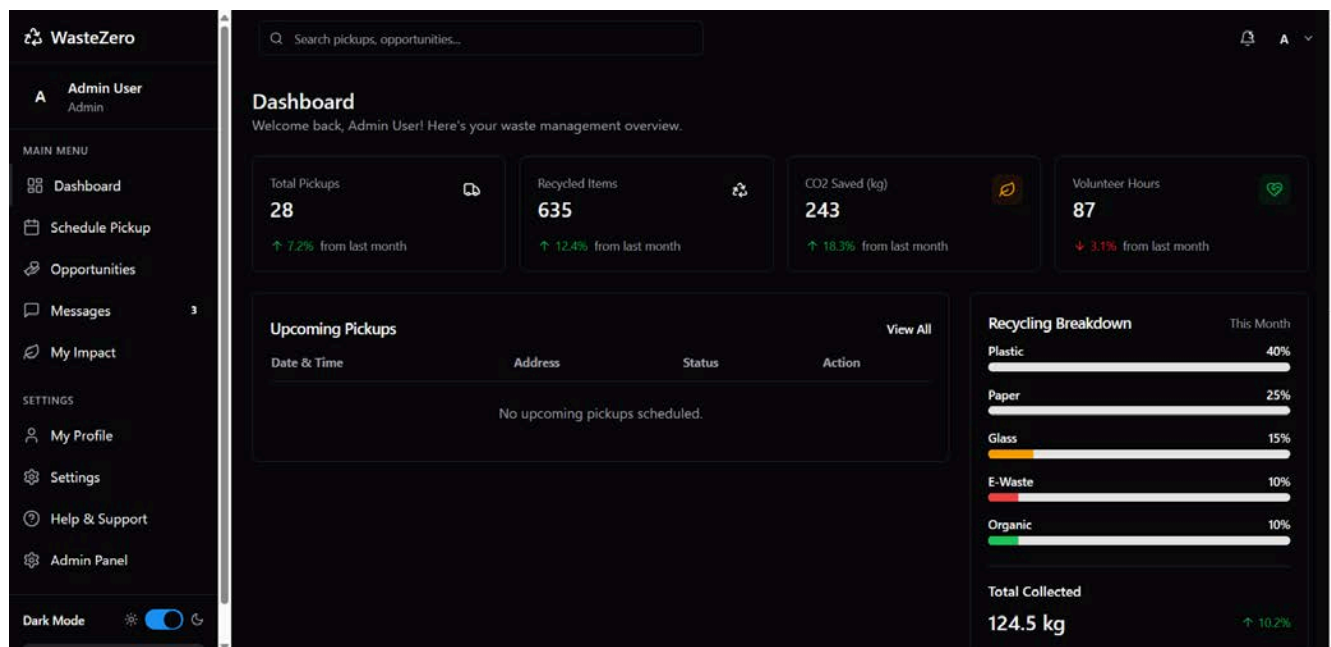
Please provide any additional details that might help our pickup team.

Previous Step

Schedule Pickup



Dark mode:



Responsive screens:
Login and Register:

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Easily arrange waste collection

Track Impact

Monitor your environmental contribution

Volunteer

Join recycling initiatives

LoginRegister

Login to your account

Enter your credentials to access your account

Username

Your username

Password

Your password

Login

LoginRegister

Create a new account

Fill in your details to join WasteZero

Full Name

Your full name

Email

Your email

Username

Choose a username

Password

Create a password

Confirm Password

Confirm your password

Role

Volunteer

Location (Optional)

Your city or area

Create Account

Dashboard screens:

Search pickups, opportunitie

A

Dashboard

Welcome back, Admin User! Here's your waste management overview.

Total Pickups

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Dashboard

Schedule

Volunteer

Chat

Profile

Search pickups, opportunitie

A

Upcoming Pickups

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Date & Time

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Status

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Volunteer Opportunities

View All



Dashboard

Schedule

Volunteer

Chat

Profile

Opportunities screens:

Search pickups, opportunitie

A

Volunteer Opportunities

View All



Beach Cleanup Drive

Open

Join us for a day of cleaning up the shoreline and protecting marine life

2025-06-20

Brighton Beach, Boston

Apply Now



Recycling Workshop

Open

Teach community members about proper recycling techniques

2025-06-15

Community Center, Seattle

Dashboard

Schedule

Volunteer

Chat

Profile

Search pickups, opportunitie

A

Apply Now



School Education Program

Open

Visit local schools to raise awareness about waste management

2025-07-10

Various schools in Boston

Apply Now

Recent Messages

View All

A

Admin User

hello

05:41 PM

A

Admin User

Your dedication to environmental causes is tr...

05:34 PM

A

Admin User

Thank you for your active participation in our...

05:28 PM

Dashboard

Schedule

Volunteer

Chat

Profile

Schedule Pickup screens:

Search pickups, opportunitie

A

Schedule Pickup

Request waste collection and manage your pickups

Schedule New Pickup

Pickup History

Request Waste Collection

Fill in the details to schedule a pickup for your recyclable waste

Address

Enter your street address

City

Enter your city

Pickup Date

mm/dd/yyyy

Preferred Time Slot

Select a time slot

Next Step

Dashboard

Schedule

Volunteer

Chat

Profile

Search pickups, opportunitie

A

Schedule Pickup

Request waste collection and manage your pickups

Schedule New Pickup

Pickup History

Your Pickup History

View and manage all your scheduled pickups

You haven't scheduled any pickups yet.

Schedule your first pickup

Dashboard

Schedule

Volunteer

Chat

Profile

Volunteer Opportunities screens:

Search pickups, opportunitie

A

Volunteer Opportunities

Browse and join recycling and waste management initiatives

+

Create Opportunity

Q

Search opportunities...

All Statuses



Beach Cleanup Drive

Open

Join us for a day of cleaning up the shoreline and protecting marine life

2025-06-20

Brighton Beach, Boston

4 hours

View Details



Dashboard

Schedule

Volunteer

Chat

Profile

Search pickups, opportunitie

A

Description

Provide details about the volunteer opportunity

Date

mm/dd/yyyy

Duration

E.g., 2 hours, 3-4 hours

Location

E.g., Ocean Beach, San Francisco

Required Skills

E.g., Teamwork, First Aid, Environmental Know

Separate skills with commas

Image URL (Optional)

URL to an image for this opportunity

Provide a URL to an image that represents this volunteering opportunity

Cancel

Create Opportunity

Dashboard

Schedule

Volunteer

Chat

Profile

Create Opportunities and messages screen:

Search pickups, opportunitie

A

← Back to Opportunities

Create New Opportunity

Post a volunteer opportunity for waste management and recycling

Opportunity Details

Title

E.g., Beach Cleanup Drive

Description

Provide details about the volunteer opportunity

Date

mm/dd/yyyy

Duration

E.g., 2 hours, 3-4 hours

Location

E.g., Ocean Beach, San Francisco

Dashboard

Schedule

Volunteer

Chat

Profile

Search pickups, opportunitie

A

Messages

Chat with volunteers, NGOs, and waste management partners

Search messages...

U

User 5

5/7/2025

You: hello

U

User 4

5/7/2025

You: Thank you for your active participation in o...

No conversation selected

Select a conversation from the left to start chatting

Dashboard

Schedule

Volunteer

Chat

Profile

My profile screen:

Search pickups, opportunitie

A

My Profile

Manage your account information and settings

Profile

Password

Personal Information

Update your personal information and profile details

Full Name

Admin User

Email

admin@wastezero.com

This is the email address used for account notifications.

Location

San Francisco, CA

This helps match you with nearby opportunities.

Skills

management, programming, sustainability

Separate skills with commas.

Bio

System administrator

Dashboard

Schedule

Volunteer

Chat

Profile

WasteZero

A

Admin User

Admin

MAIN MENU

Dashboard

Schedule Pickup

Opportunities

Messages

My Impact

SETTINGS

My Profile

Settings

Help & Support

Admin Panel

Dark Mode

Sign Out